

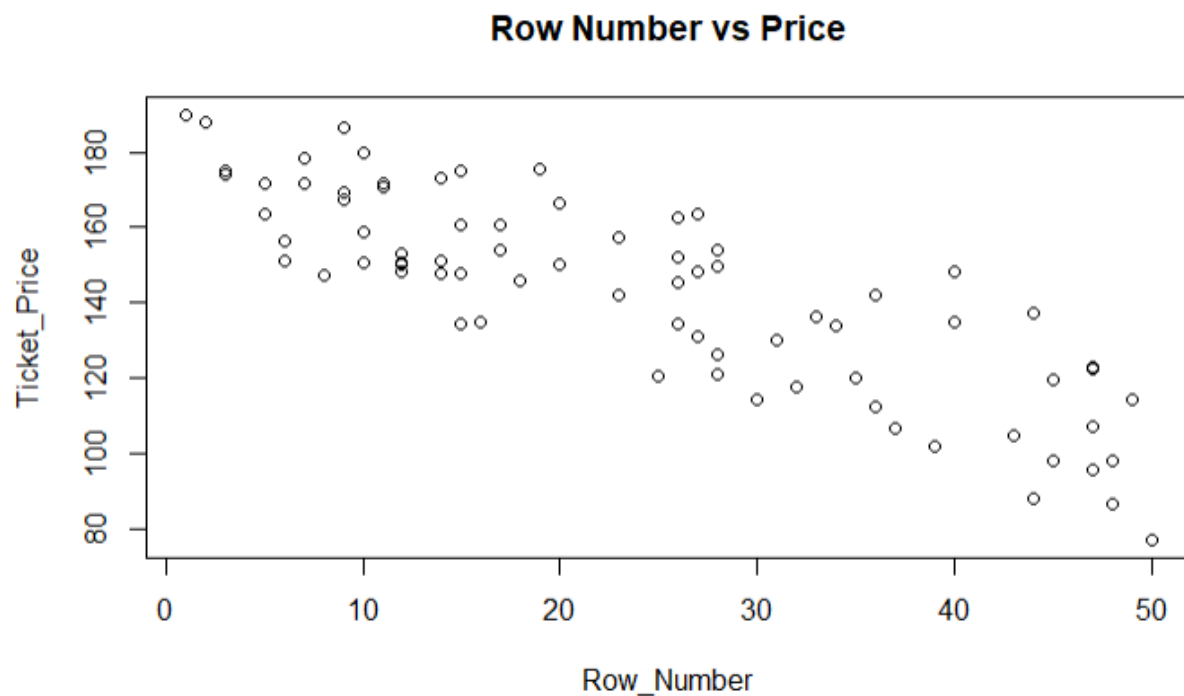
Understand the Problem and Data:

This report aims to explain the price of tickets (dollars) using specific variables from our dataset

“Ticket_Sales.csv”. These variables would include...

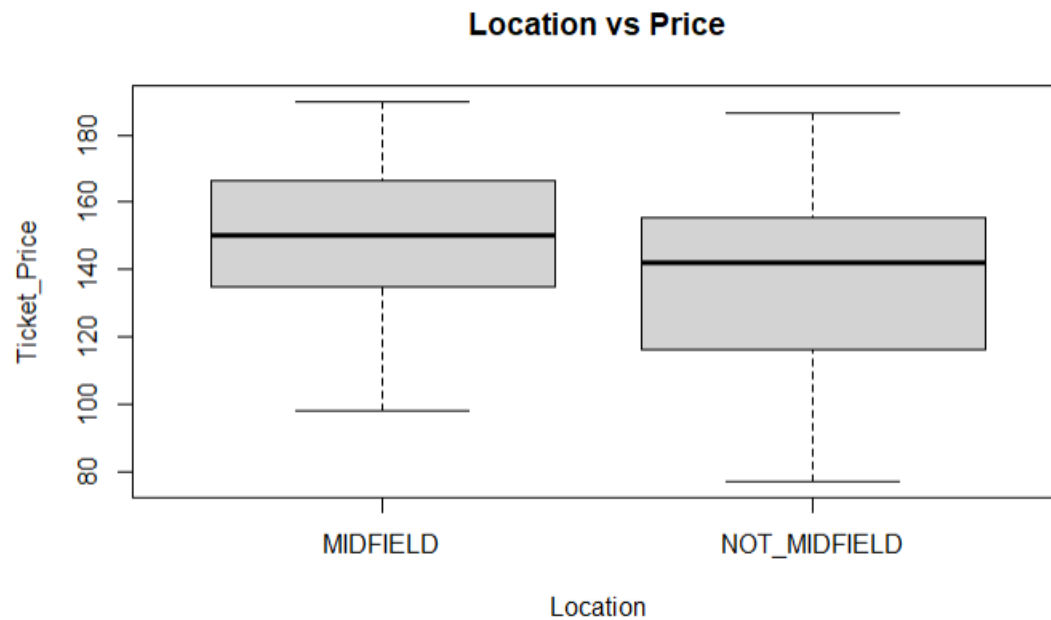
1. Row_Number (Seat and Row from field)
2. Location (Section of seat; midfield or not midfield)
3. Aisle (is on the aisle or not)
4. Purchase_Before_Event (Number of days before event that the ticket was purchased)
5. Game_Day (Day of the week for the game; either Tuesday or Thursday)

We won't use Ticket_ID since it is unique for each transaction. Statistical analysis aims to identify patterns and trends from the data by collecting and interpreting information.

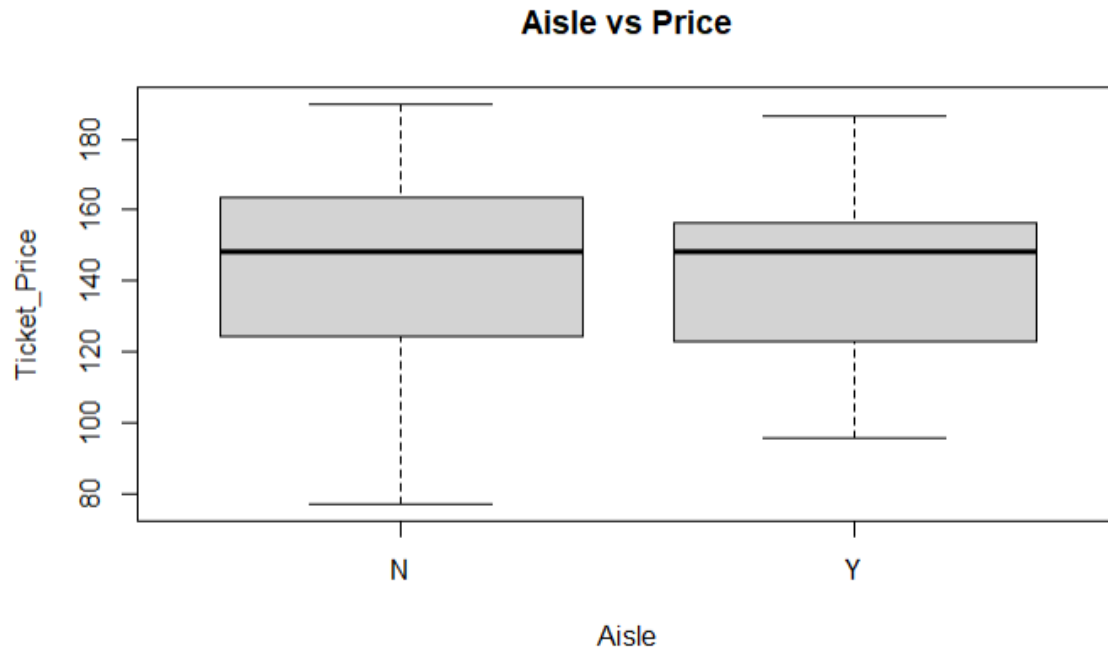


As shown above, there is a negative linear relationship between the price of the ticket and the row number, meaning that on average, ticket prices will most likely be more expensive the closer they are to

the field (lower row number). The variability also seems to be kept constant through the entirety of the graph.



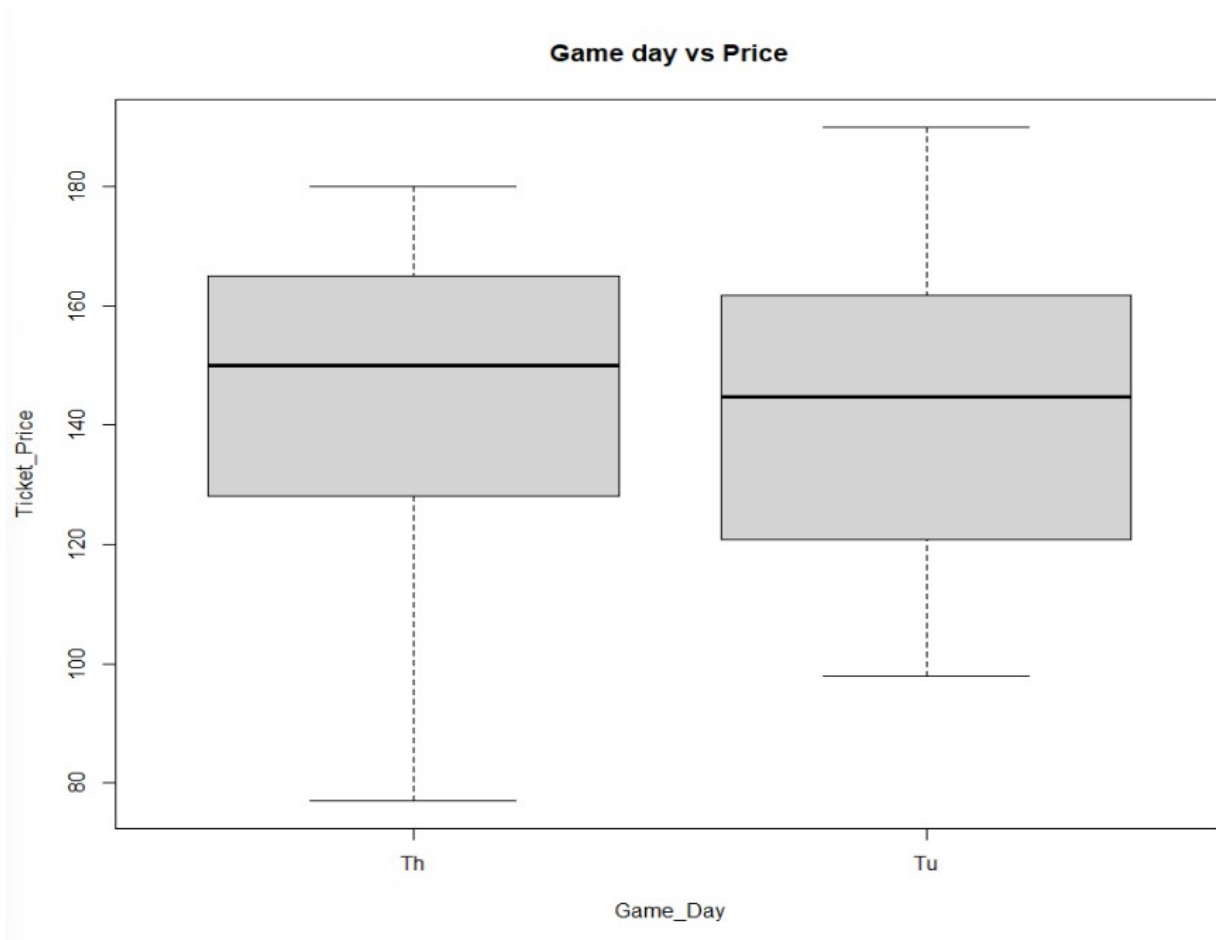
As seen from the boxplot above, tickets for seats in the midfield, on average, are more expensive than tickets that are seats that aren't in the midfield. Another thing to notice here would be the difference in ranges; a ticket for a midfield seat can cost between \$100 to over \$180 dollars, compared to a ticket outside of the midfield ranging from \$80 to a little over \$180 dollars.



As seen from the boxplot above, the median for both tickets that are on/ not on the aisle are about the same, around \$150. However, the IQR and range for ticket prices that are on the aisle are smaller than the prices for tickets that are not.



As shown from the graph above, there doesn't seem to be a pattern/ trend that we can easily pick out. There doesn't seem to be any relationship, positive or linear. However, the variance appears to be constant throughout the entire graph.



As shown from the boxplot above, the ticket prices for games on Thursday, on average, are more expensive than tickets for games on Tuesday. A thing to note would be the differences in ranges; tickets on Thursday can sell for less (\$80-\$180) compared to Tuesday (\$100-\$180+).

Model Specification:

Possible selection types

Best subset selection: 32 regression models (2^5)

Forward Stepwise selection: 16 (1 + 5(5 + 1)/2)

Backwards Stepwise selection: 16 (same as forward)

For the sake of time, we will be using Forward Stepwise selection, which means we will start with a model with no predictor variables, then we will add 1 variable to the model and pick the model with the best r^2 value. Then we will build upon that best model with another predictor variable and rinse repeat that process until all variables are in the model. Then, after we have all the best models for n amount of predictor variables, we then pick the best of the best. In other terms, here is a summary in terms of equations:

- $y = \beta_0 + \epsilon$
- $y = \beta_0 + \beta_{PT}x_{PT} + \epsilon$
- $y = \beta_0 + \beta_{PCI}x_{PCI} + \epsilon$
- $y = \beta_0 + \beta_{MH}x_{MH} + \epsilon$
- $y = \beta_0 + \beta_{PD}x_{PD} + \epsilon$
- $y = \beta_0 + \beta_{PD}x_{PD} + \beta_{PT}x_{PT} + \epsilon$
- $y = \beta_0 + \beta_{PD}x_{PD} + \beta_{PCI}x_{PCI} + \epsilon$
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- $y = \beta_0 + \beta_{PD}x_{PD} + \beta_{PCI}x_{PCI} + \beta_{MH}x_{MH} + \epsilon$
- $y = \beta_0 + \beta_{MH}x_{PD} + \beta_{PT}x_{MH} + \epsilon$

Every single coefficient that isn't B, x, or e represents a different predictor variable.

Model Estimation:

A model with 0 variables would just be the mean of the Ticket_Price

143.158

Models with 1 Variable:

Predictor Variable	Multiple R-Squared Value
Row Number	.713
Location	.07
Aisle	.0002
Purchase Before Event	.046
Game Day	.0001

Showcase the multiple R-Squared value since it's only one predictor variable and not multiple

Proceeding with 2 Predictor variables that INCLUDE Row Number

P Variable + Row Number	Adjusted R Squared
Location	.794
Aisle	.712
Purchase Before Event	.706
Game Day	.706

*We will use Adjusted R Squared from now on, since there are more than 1 predictor variables

Proceeding with 3 P variables that include Row Number + Aisle

P Variable + Row Num + Location	Adjusted R Squared
Aisle	.804
Purchase Before Event	.791
Game Day	.793

Proceeding with 4 P Variables that include all of the above + Aisle

P Variable + RowNum + Loc + Aisle	Adjusted R Squared
Purchase Before Event	.801
Game Day	.802

Last, we have the model that uses every single predictor variable. This gives us an adjusted R squared value of .800

With all of this in mind, we are going to proceed with the model that has 3 predictor variables, Row Number, Location, and Aisle, since it has the highest R Squared value of .804

Model Assessment:

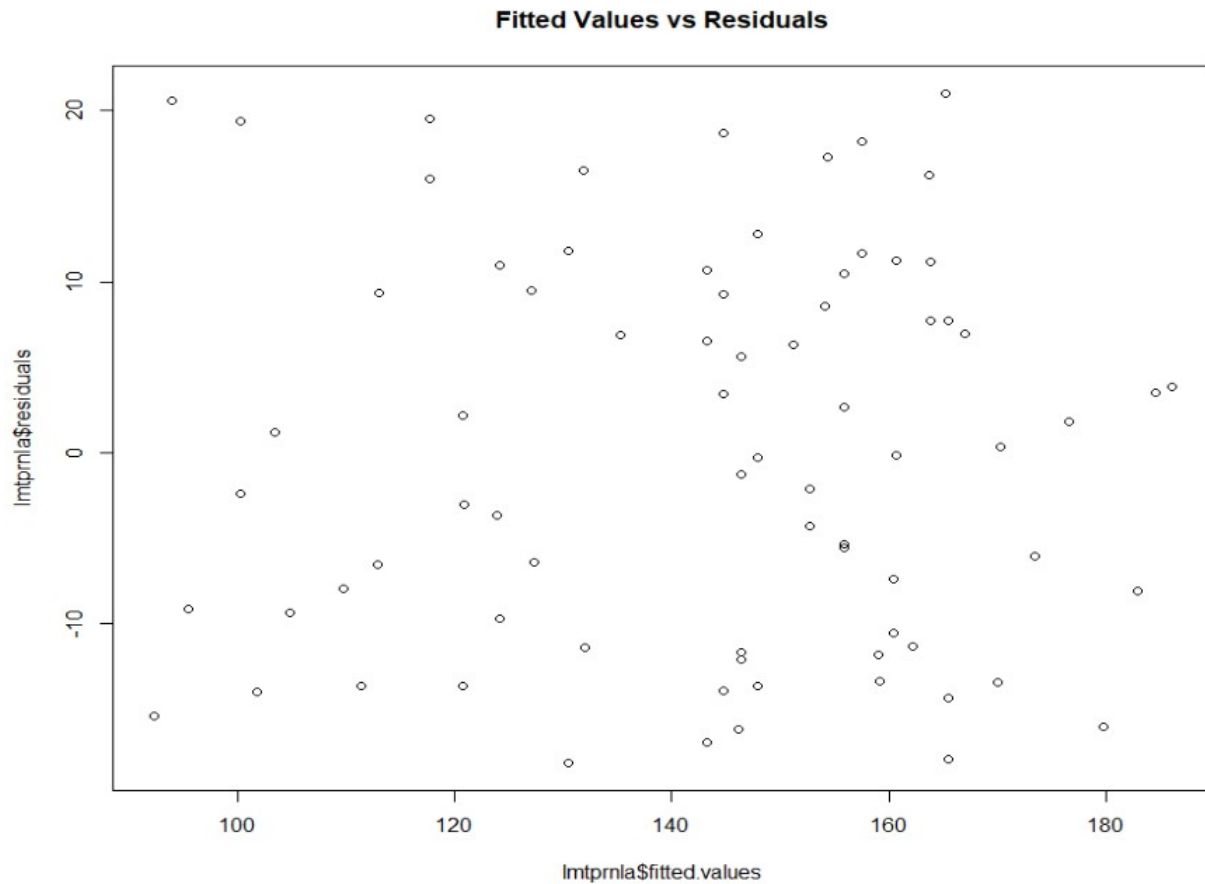
Now, we will use 4-fold cross-validation to choose the best overall model. This means that we will divide our data into 4 groups, and each of these groups will have 19 observations (since we have 76 in total), in order to fit 4 different models. Each model will basically use 1 group as the test set and the other 3 sets as a training set.

Here is a table listing off each of our models and the cross-validation errors that go along with them

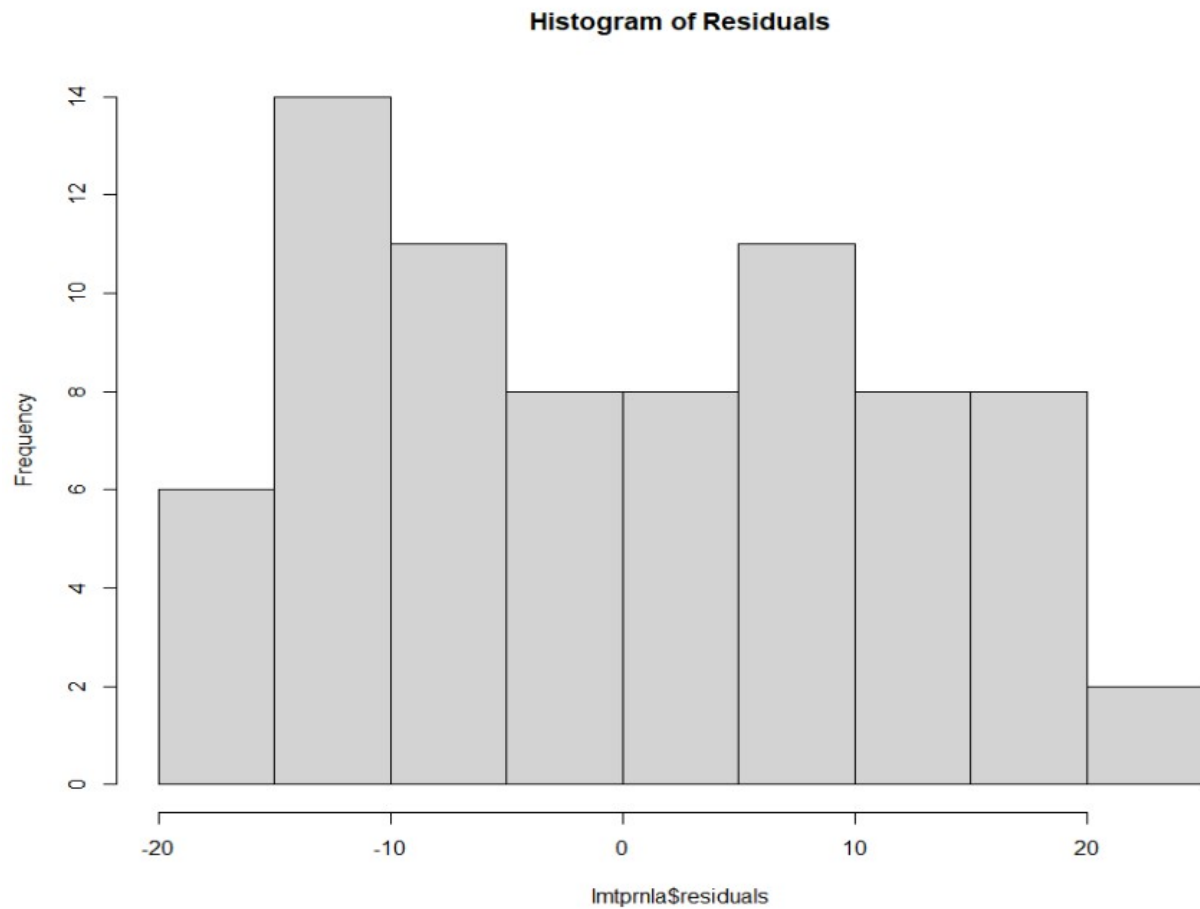
# of predictor Variables in model	Cross Validation Error
0	707.783
1	207.671
2	149.411
3	145.660
4	148.756
5	152.790

After scanning through each of the models validation errors, we can easily choose our model with 3 predictor variables since it has the lowest error compared to every other model, which also goes along with what we chose in our Model Estimation tab using the R Squared to make our decision.

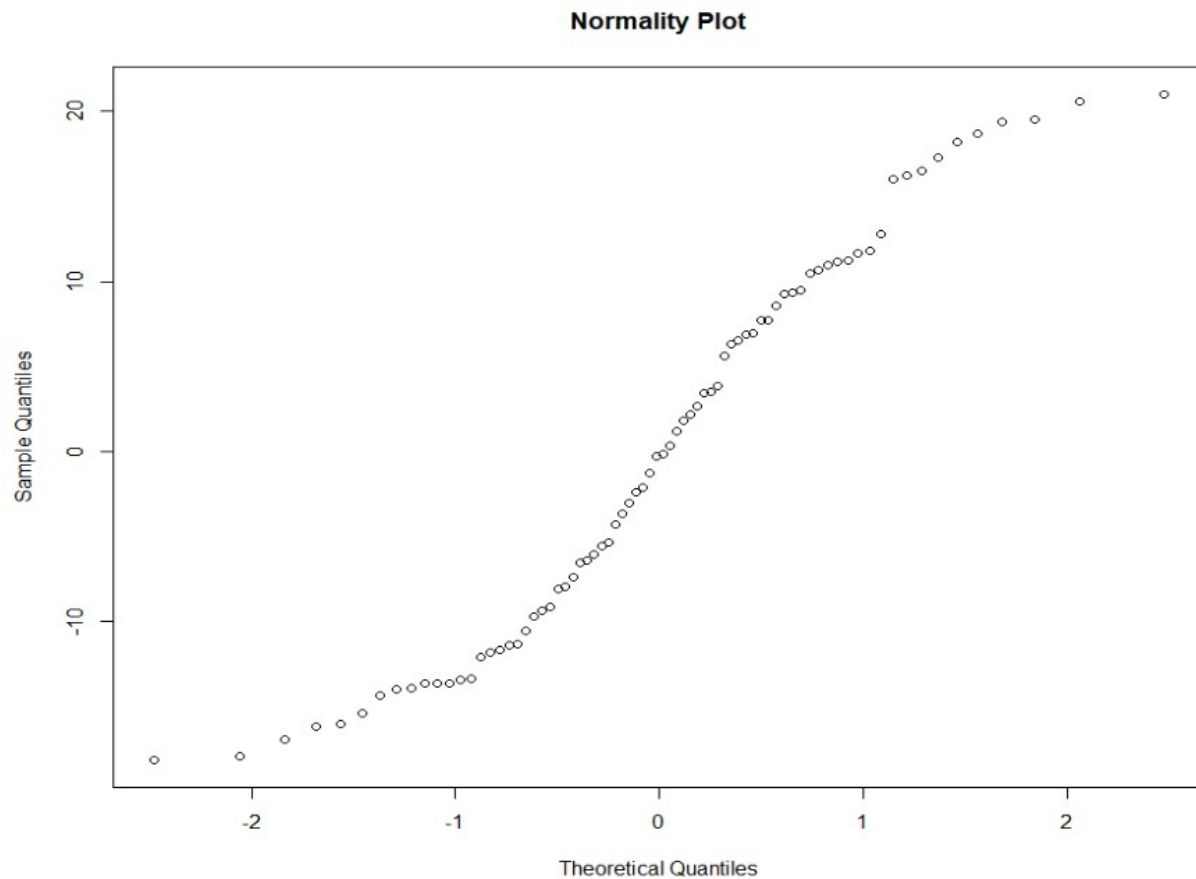
Diagnostic Checking



The residuals do not seem to be centered around zero across fitted values. There doesn't seem to be any changes in variance across fitted values, and there is also a lack of any pattern/ trend.



When creating a histogram of the model, it is clear that the residuals do not follow a normal distribution, since there seems to be 2 peaks instead of one.



A normality plot reveals that the residuals do not follow a straight line due to lots of deviation.

```
Shapiro-Wilk normality test
data: lmtprn1a$residuals
W = 0.94773, p-value = 0.003475
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A Shapiro-Wilk test gives us a P-value of .003, which is less than our alpha .05, meaning we reject the null hypothesis that the residuals follow a normal distribution.

With all this in mind, I would be skeptical to deploy this model since there is concrete evidence that the residuals are not normally distributed, since normally distributed residuals ensure more reliable p values.

In order to find an appropriate model to explain the ticket price, I would proceed with the transformation

of our data in order to make our relationship between Ticket Prices and our predictor variables much more appropriate.